

14 – Forms

A way to send information from the client to the server

Jérôme Landré – jerome.landre@ju.se



HOW ARE YOU?



HUR ÄR DET MED DIG? HUR ÄR LAGET?



ÇA VA?



I AM FINE/OK/GOOD.



DET ÄR LUNGT. ALLT ÄR FINT/BRA.



ÇA VA.

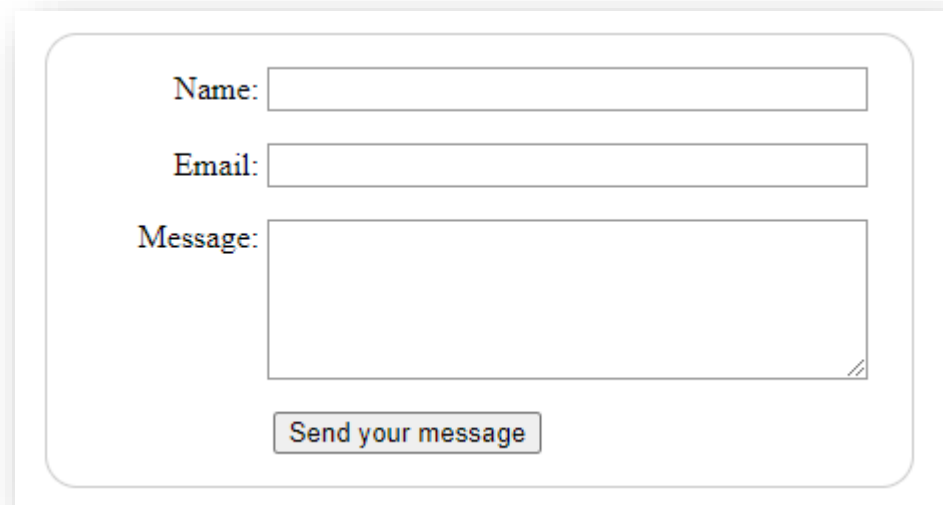
OUTLINE

- 1) Forms
- 2) Forms within Node & Express
- 3) Forms Summary



1) FORMS

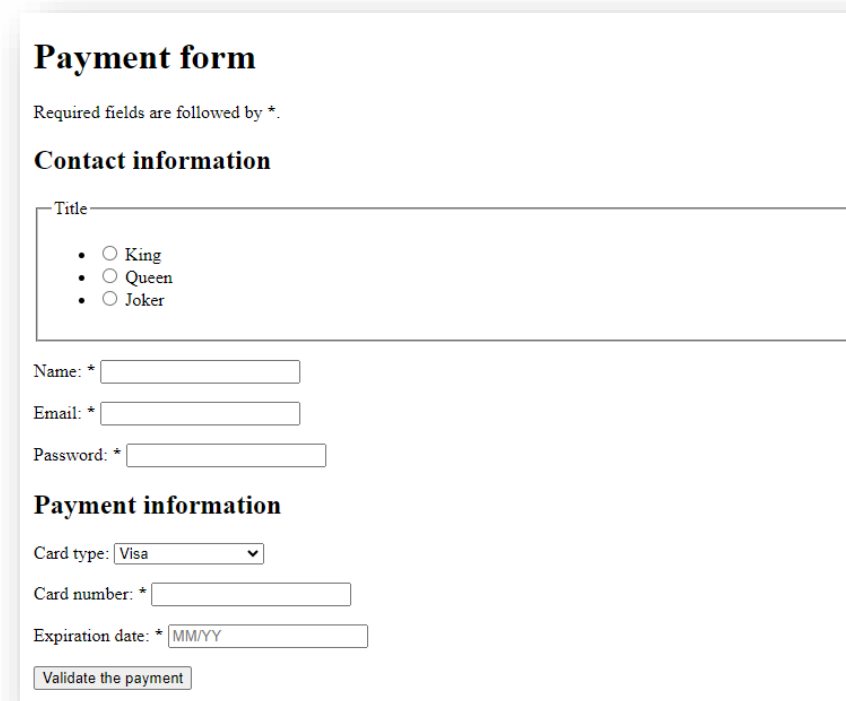
- Forms are a part of HTML to create user interface to send information from the client to the server



A simple contact form with the following fields:

- Name:
- Email:
- Message:
-

Source: https://developer.mozilla.org/en-US/docs/Learn/Forms/Your_first_form



Payment form

Required fields are followed by *.

Contact information

Title

- ☐ King
- ☐ Queen
- ☐ Joker

Name: *

Email: *

Password: *

Payment information

Card type:

Card number: *

Expiration date: *

Source: https://developer.mozilla.org/en-US/docs/Learn/Forms/How_to_structure_a_web_form

1) FORMS

- In HTML, you can create forms with the <form> tag:

https://www.w3schools.com/html/html_forms.asp

https://developer.mozilla.org/en-US/docs/Learn/Forms/Basic_native_form_controls

https://developer.mozilla.org/en-US/docs/Learn/Forms/Other_form_controls

- You can control form fields with some kind of client-side validation

https://developer.mozilla.org/en-US/docs/Learn/Forms/Form_validation

- But remember, you **MUST** check the validity of the entries of your forms fields on the client and/or on the server: **don't trust form users!**

1) FORMS

- A form has two default attributes: **method** and **action**
- **method** specifies the method to send information:
 - **GET**: the information are sent inside the URL has
URL?**name1=value1&name2=value2&name2=value3**
 - **POST**: the information are sent in the body of the request
- **action** specifies the route or path to send the information to
- A form has **one or more fields** (text, textarea, radio, checkbox, select/options, date, file, hidden...)
- A form has a "**submit**" button **inside** it (don't put the button outside the form)

1) FORMS

<https://www.tutorialrepublic.com/html-tutorial/html-forms.php>

```
views > login.handlebars > ...
1  <div id="main">
2    <div>
3      <h1>Log in</h1>
4      <div>
5        <form action="/login" method="post" id="formlogin">
6          <fieldset>
7            <legend>Login information</legend>
8            <label>Username: <input type="text" name="un"></label>
9            <br />
10           <label>Password: <input type="password" name="pw"></label>
11          </fieldset>
12          <input type="submit" value="Login">
13        </form>
14      </div>
15    </div>
16  </div>
17
```

Log in

Login information

Username:

Password:

Login

2) FORMS WITHIN NODE & EXPRESS

- GET method:

```
// defines route "/login-get"  
app.get('/login-get', function(request, response){  
  response.render('login-get.handlebars')  
})
```

```
views > login-get.handlebars > ...  
1  <div id="main">  
2    <div>  
3      <h1>Log in</h1>  
4      <div>  
5        <form action="/login-test" method="get" id="formlogin">  
6          <fieldset>  
7            <legend>Login information</legend>  
8            <label>Username: <input type="text" name="un"></label>  
9            <br />  
10           <label>Password: <input type="password" name="pw"></label>  
11          </fieldset>  
12          <input type="submit" value="Login">  
13        </form>  
14      </div>  
15    </div>  
16  </div>  
17  |
```

2) FORMS WITHIN NODE & EXPRESS

- GET method: the information is passed in the URL => no **password** please!

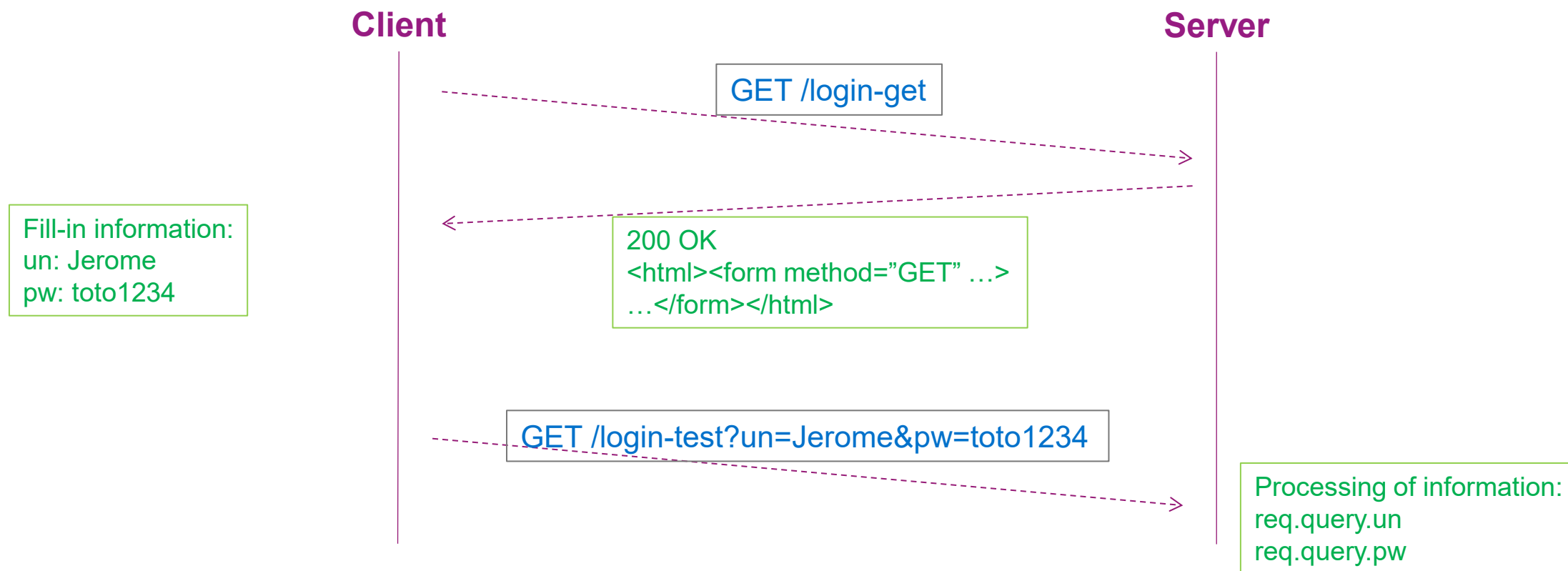
```
PS C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl> node app.js
Server running and listening on port 8080...
URL: /login-test?un=J%C3%A9r%C3%B4me&pw=toto1234
Received login/password: Jérôme/toto1234
█
```



```
app.get('/login-test', (req,res) => {
  // prints the oute
  console.log("URL: ", req.url)
  // gets GET results from the URL
  const login = req.query.un
  const password = req.query.pw
  console.log("Received login/password: "+login+'/'+password )
  res.end()
})
```

2) FORMS WITHIN NODE & EXPRESS

- GET method:



2) FORMS WITHIN NODE & EXPRESS

- POST method:

```
// defines route "/login-post"  
app.get('/login-post', function(request, response){  
  response.render('login-post.handlebars')  
})
```

```
views > login-post.handlebars > ...  
1  <div id="main">  
2    <div>  
3      <h1>Log in</h1>  
4      <div>  
5        <form action="/login-test" method="post" id="formlogin">  
6          <fieldset>  
7            <legend>Login information</legend>  
8            <label>Username: <input type="text" name="un"></label>  
9            <br />  
10           <label>Password: <input type="password" name="pw"></label>  
11          </fieldset>  
12          <input type="submit" value="Login">  
13        </form>  
14      </div>  
15    </div>  
16  </div>  
17
```

2) FORMS WITHIN NODE & EXPRESS

- POST method: the information is passed in the request => yes for **password!**

```
PS C:\Users\lanjer\Desktop\web-dev-fun\portfolio-jl> node app.js
Server running and listening on port 8080...
URL: /login-test
POST data: {"un":"Jérôme","pw":"toto1234"}
Received login/password: Jérôme/toto1234
```

`npm install body-parser`

```
const bodyParser = require('body-parser') // line to put at the top of the file...
// ... where the other require are

// lines to put just after the app = express()
app.use(bodyParser.urlencoded({ extended: false }))
app.use(bodyParser.json())

// form in POST
app.post('/login-test', (req,res) => {
  // prints the route
  console.log("URL: ", req.url)
  // prints the received POST data
  var post_data = JSON.stringify(req.body)
  console.log("POST data: ", post_data)
  // gets GET results from the URL
  const login = req.body.un
  const password = req.body.pw
  console.log("Received login/password: "+login+'/'+password )
  res.end()
})
```

Home Projects About Contact Login

Log in

Login information

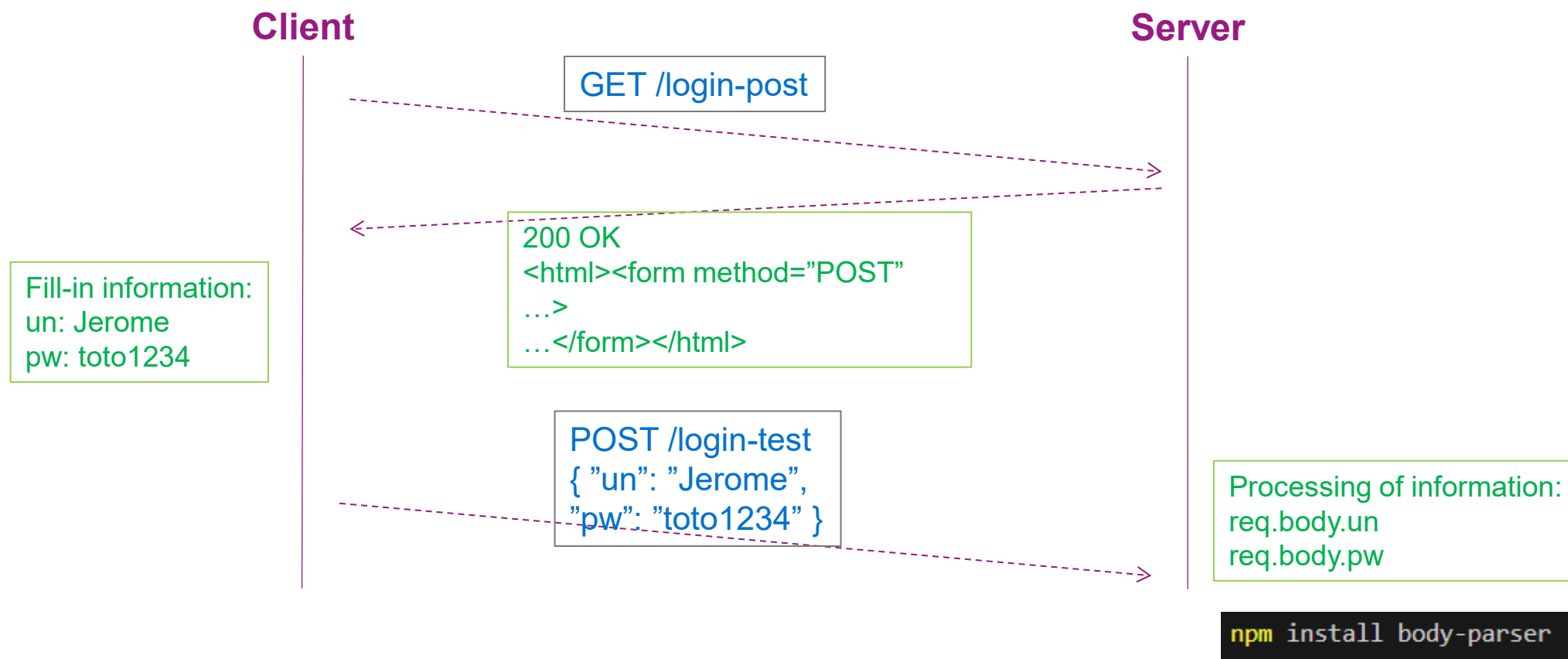
Username: Jérôme

Password:

Login

2) FORMS WITHIN NODE & EXPRESS

- POST method:



3) FORMS SUMMARY

- A form has two default attributes: **method** and **action**
 - **method** specifies the method to send information:
 - **GET**: the information are sent inside the URL has
URL?**name1=value1&name2=value2&name2=value3**
 - **POST**: the information are sent in the body of the request
 - **action** specifies the route or path to send the information to
- **GET**: data is passed into the URL => **no password!**
- **POST**: data is passed in the body of the request: **OK for password!**

ANY QUESTIONS?